

ECE 332 — Homework 2: Generated Answers from Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

Problem 1 — Plane Wave E and H with Phase Constant

10 pts

Setup: 1 GHz wave traveling $+\hat{y}$ in lossless nonmagnetic medium with $\epsilon_r = 9$. $\mathbf{E}$ along $\hat{x}$, peak 6 V/m, $|E| = 4$ V/m at $t = 0$, $y = 2$ cm.

PLANE WAVE PARAMETERS (LOSSLESS)	LOSSLESS PLANE WAVE — E AND H
Angular frequency ω: rad/s; f: Hz $\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$ Phase velocity m/s; <i>nonmagnetic</i>: $\mu_r = 1$ $u_p = \frac{c}{\sqrt{\epsilon_r}} \quad c = 3 \times 10^8 \text{ m/s}$ Wavenumber k rad/m $k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$ Wavelength & impedance m & Ω $\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \text{ }\Omega$ <i>Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0)\hat{e}$.</i>	General E $\mathbf{E} = E_0 \cos(\omega t - \mathbf{k} \cdot \mathbf{r} + \phi_0) \hat{e}$ $\hat{k}$ = direction of travel (unit vector), e.g. $+\hat{y} \Rightarrow \hat{k} = \hat{y}$. k = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\epsilon_r}/c = 2\pi/\lambda$. ϕ_0 = initial phase constant (rad): shift set by given condition at known t, position. $\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$. $\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k} \cdot \mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k} \cdot \mathbf{r} = y$. H from E $\hat{k}, \hat{E}, \hat{H}$ form right-hand set $\mathbf{H} = \frac{1}{\eta} (\hat{k} \times \mathbf{E})$ Find ϕ_0 given $E = E_1$ at $t = 0$, position r_0 $E_0 \cos(-kr_0 + \phi_0) = E_1 \Rightarrow \phi_0 = \cos^{-1}\left(\frac{E_1}{E_0}\right) + kr_0$ $\cos^{-1}$ gives $[0, \pi]$; take ϕ_0 in that range.

Parameters from the cheat sheet order $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$:

$$\omega = 2\pi f = 2\pi \times 10^9 \text{ rad/s}, \quad k = \frac{\omega\sqrt{\epsilon_r}}{c} = \frac{2\pi \times 10^9 \sqrt{9}}{3 \times 10^8} = 20\pi \text{ rad/m}$$

General E form (wave in $+\hat{y}$ so argument is $\omega t - ky$):

$$\mathbf{E}(y, t) = 6 \cos(\omega t - ky + \phi_0) \hat{x} \text{ V/m}$$

Find ϕ_0 from the given condition $|E| = 4$ at $t = 0$, $y = 0.02$ m:

$$4 = 6 \cos(-0.4\pi + \phi_0) \Rightarrow -0.4\pi + \phi_0 = \cos^{-1}(2/3) = 0.84 \text{ rad}$$

$$\phi_0 = 2.1 \text{ rad}$$

$$\mathbf{E}(y, t) = 6 \cos((2\pi \times 10^9)t - (20\pi)y + 2.1) \hat{x} \text{ V/m}$$

Magnetic field via $\mathbf{H} = \frac{1}{\eta}(\hat{k} \times \mathbf{E})$:

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} = \frac{377}{3} \approx 125.7 \text{ }\Omega, \quad \hat{y} \times \hat{x} = -\hat{z}$$

$$\mathbf{H}(y, t) = -\frac{6}{125.7} \cos((2\pi \times 10^9)t - (20\pi)y + 2.1) \hat{z} \text{ A/m}$$

Problem 2 — f, u_p, λ, k, η and $\mathbf{H}$

12 pts

Setup: $\mathbf{E} = 20 \cos((6\pi \times 10^9)t - kz) \hat{y}$ V/m, lossless nonmagnetic, $\epsilon_r = 2.56$.

PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency ω : rad/s; f : Hz

$$\omega = 2\pi f \quad f = \frac{\omega}{2\pi}$$

Phase velocity m/s ; nonmagnetic: $\mu_r = 1$

$$u_p = \frac{c}{\sqrt{\epsilon_r}} \quad c = 3 \times 10^8 \text{ m/s}$$

Wavenumber k rad/m

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\epsilon_r}}{c}$$

Wavelength & impedance m & Ω

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k} \text{ (m)} \quad \eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}, \quad \eta_0 = 377 \Omega$$

Order: $\omega \rightarrow u_p \rightarrow k \rightarrow \lambda \rightarrow \eta$. Base: $\mathbf{E} = E_0 \cos(\omega t \pm kx + \phi_0) \hat{e}$.

 $\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k} \times \hat{k} = \mathbf{0}$; any $\hat{a} \times \hat{a} = \mathbf{0}$.

Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

(a) Apply the parameter chain in order:

$$f = \omega/(2\pi) = 3 \times 10^9 \text{ Hz}$$

$$u_p = c/\sqrt{\epsilon_r} = 3 \times 10^8 / \sqrt{2.56} = 1.875 \times 10^8 \text{ m/s}$$

$$\lambda = u_p/f = 6.25 \text{ cm}$$

$$k = 2\pi/\lambda = 32\pi \approx 100.5 \text{ rad/m}$$

$$\eta = \eta_0/\sqrt{\epsilon_r} = 377/\sqrt{2.56} = 235.6 \Omega$$

(b) $\mathbf{H}$ direction from the $\hat{k} \times \hat{E}$ table:

Wave in $+\hat{z}$, $\mathbf{E}$ along $\hat{y}$, so $\hat{z} \times \hat{y} = -\hat{x}$.

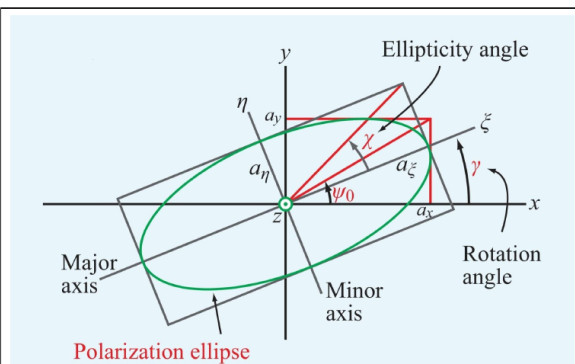
$$\mathbf{H}(z, t) = -\frac{20}{235.6} \cos((6\pi \times 10^9)t - (32\pi)z) \hat{x} = -0.0849 \cos(\cdot) \hat{x} \text{ A/m}$$

Problem 3 — Polarization Analysis (4 cases)

16 pts

Setup: $\mathbf{E}(z, t) = a_x \cos(\omega t - kz)\hat{x} + a_y \cos(\omega t - kz + \delta)\hat{y}$.

POLARIZATION — GENERAL WAVE



$$\mathbf{E} = a_x \cos(\omega t - kz) \hat{x} + a_y \cos(\omega t - kz + \delta) \hat{y}$$

a_x, a_y : bounding box half-widths (V/m); δ : phase diff (rad)

$$\delta = 0: \mathbf{E} = (a_x \hat{x} + a_y \hat{y}) \cos(\omega t - kz) \text{ (linear)}$$
$$\delta = \pi: \mathbf{E} = (a_x \hat{x} - a_y \hat{y}) \cos(\omega t - kz) \text{ (linear)}$$

Auxiliary angle $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$

Rotation angle γ (major axis from $\hat{x}$) $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0) \cos \delta$$

If $\text{sign}(\gamma) \neq \text{sign}(\delta)$: $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$.

Ellipticity angle $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0) \sin \delta$$

Axial ratio $R = 1/\tan|\chi|$ *major axis* = $R \times$ *minor*; $R \geq 1$

QUADRANT RULE FOR $\tan(2\gamma)$

$$N = \sin(2\psi_0) \cos \delta, \quad D = \cos(2\psi_0):$$

N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	-	$180^\circ - \arctan N/D $	$\pi - \arctan N/D $
-	-	$180^\circ + \arctan N/D $	$\pi + \arctan N/D $
-	+	$360^\circ - \arctan N/D $	$2\pi - \arctan N/D $

$$\gamma = (2\gamma)/2. \quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}.$$

$$\text{Convert: } \theta_{rad} = \theta_{\circ} \times \frac{\pi}{180}; \theta_{\circ} = \theta_{rad} \times \frac{180}{\pi}.$$

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0, \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4, 0)$	LHEP	LH (CW)

CIRCULAR POLARIZATION — TIME DO- MAIN

$$a_x = a_y = a, \text{ wave in } +\hat{z}$$

RHC $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$

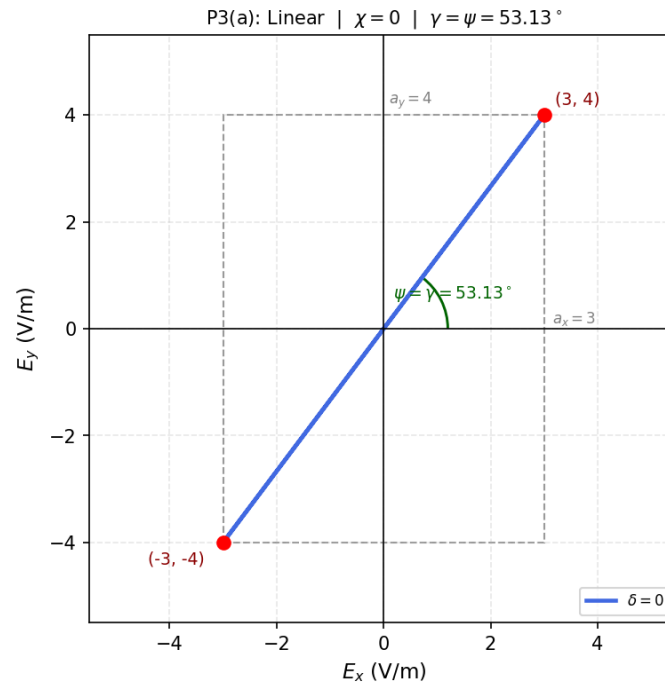
$$\mathbf{E}(z, t) = a[\cos(\omega t - kz) \hat{x} + \sin(\omega t - kz) \hat{y}]$$

LHC $\delta = +\pi/2$: $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$

$$\mathbf{E}(z, t) = a [\cos(\omega t - kz) \hat{x} - \sin(\omega t - kz) \hat{y}]$$

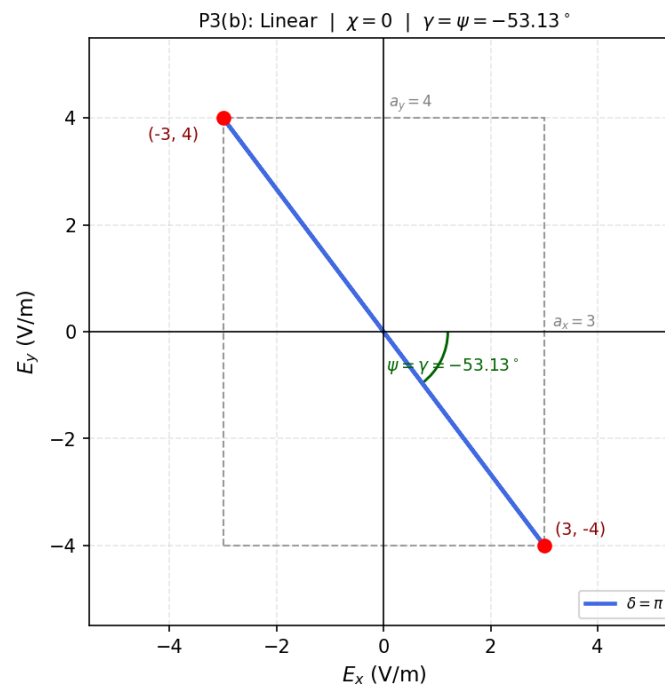
(a) $a_x = 3$, $a_y = 4$, $\delta = 0$: $\delta = 0 \Rightarrow$ linear. $R = \infty$, $\chi = 0$. Rotation angle = inclination:

$$\psi = \tan^{-1}(4/3) = 0.9273 \text{ rad} = 53.13^\circ \Rightarrow \boxed{\gamma = 53.13^\circ, \chi = 0}$$



(b) $a_x = 3$, $a_y = 4$, $\delta = \pi$: $\delta = \pi \Rightarrow$ linear again ($\cos(\theta + \pi) = -\cos \theta$, so $\mathbf{E} = (3\hat{x} - 4\hat{y}) \cos(\omega t - kz)$). $\chi = 0$.

$$\psi = \tan^{-1}(-4/3) = -0.9273 \text{ rad} = -53.13^\circ \Rightarrow \boxed{\gamma = -53.13^\circ, \chi = 0}$$



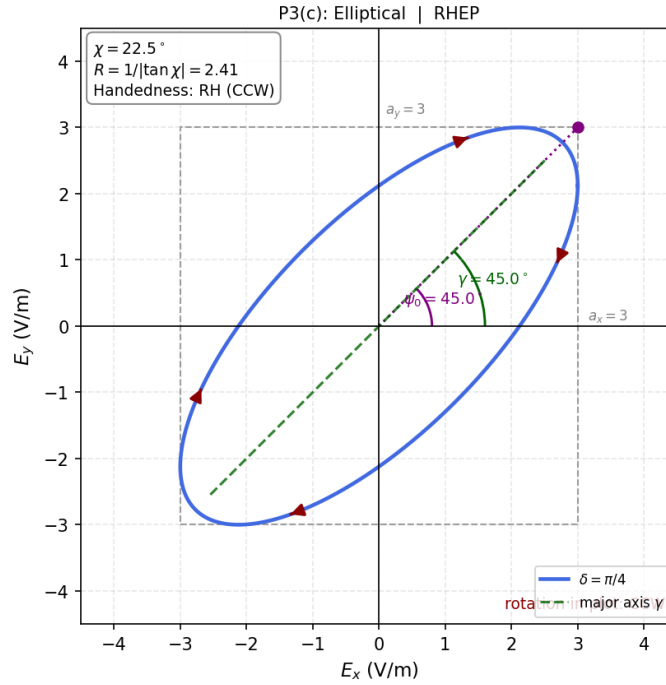
(c) $a_x = 3$, $a_y = 3$, $\delta = \pi/4$: elliptical. Auxiliary $\psi_0 = \tan^{-1}(1) = \pi/4$.

$$\tan 2\gamma = \tan(\pi/2) \cos(\pi/4) = \infty \sqrt{2}/2 \Rightarrow 2\gamma = \pi/2$$

$$\sin 2\chi = \sin(\pi/2) \sin(\pi/4) = \sqrt{2}/2 \Rightarrow 2\chi = \pi/4$$

$$\boxed{\gamma = \pi/4 = 45^\circ, \quad \chi = \pi/8 = 22.5^\circ}$$

$\chi > 0 \Rightarrow$ right-hand elliptical (RHEP).



(d) $a_x = 3$, $a_y = 4$, $\delta = -3\pi/4$: elliptical. $\psi_0 = \tan^{-1}(4/3) \approx 0.9273$, $2\psi_0 \approx 1.8546$ (106.26°).

$$\cos \delta = -\sqrt{2}/2, \quad \sin \delta = -\sqrt{2}/2$$

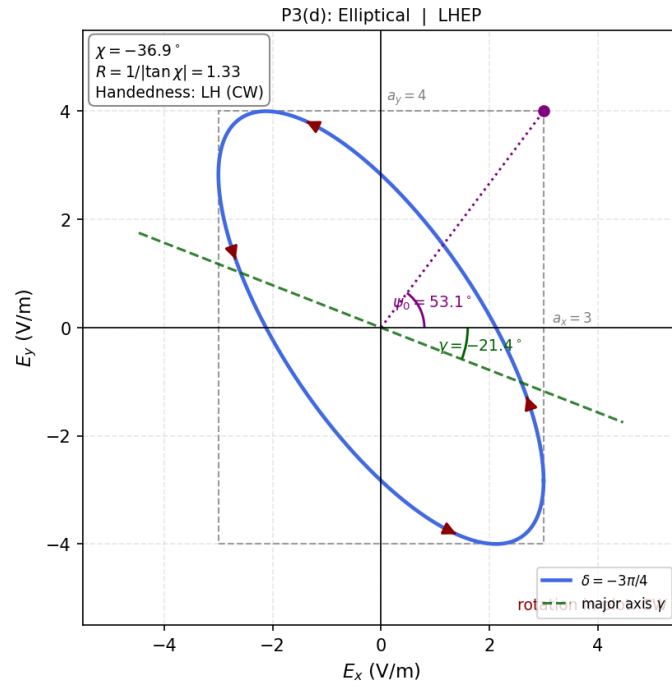
$$\tan 2\gamma = \tan(1.8546) \cdot (-\sqrt{2}/2) \approx 2.4244$$

$N = \sin(2\psi_0) \cos \delta < 0$, $D = \cos(2\psi_0) < 0$ (quadrant rule: both negative)

$$2\gamma \approx -1.1796 \text{ rad} = -73.73^\circ \Rightarrow \boxed{\gamma \approx -21.4^\circ}$$

$$\sin 2\chi = \sin(1.8546) \cdot (-\sqrt{2}/2) = -0.7462 \Rightarrow 2\chi \approx -0.7462 \text{ rad} \Rightarrow \boxed{\chi \approx -36.9^\circ}$$

$\chi < 0 \Rightarrow$ left-hand elliptical (LHEP).



Problem 4 — Linear = RHC + LHC Decomposition

8 pts

Setup: Show $\tilde{\mathbf{E}} = a_x e^{-jkz} \hat{x}$ can be written as $\tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L$.

CIRCULAR POLARIZATION — TIME DOMAIN

$a_x = a_y = a$, wave in $+\hat{z}$

RHC $\delta = -\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} - j\hat{y})e^{-jkz}$

$$\mathbf{E}(z, t) = a [\cos(\omega t - kz) \hat{x} + \sin(\omega t - kz) \hat{y}]$$

LHC $\delta = +\pi/2$; $\tilde{\mathbf{E}} = a(\hat{x} + j\hat{y})e^{-jkz}$

$$\mathbf{E}(z, t) = a [\cos(\omega t - kz) \hat{x} - \sin(\omega t - kz) \hat{y}]$$

From the cheat sheet definitions:

$$\tilde{\mathbf{E}}_R = a_R(\hat{x} - j\hat{y})e^{-jkz}, \quad \tilde{\mathbf{E}}_L = a_L(\hat{x} + j\hat{y})e^{-jkz}$$

Try $a_R = a_L = a_x/2$:

$$\begin{aligned} \tilde{\mathbf{E}}_R + \tilde{\mathbf{E}}_L &= [(a_R + a_L)\hat{x} + j(-a_R + a_L)\hat{y}]e^{-jkz} \\ &= [a_x \hat{x} + j \cdot 0 \cdot \hat{y}]e^{-jkz} = a_x e^{-jkz} \hat{x} = \tilde{\mathbf{E}} \quad \checkmark \end{aligned}$$

$$a_R = a_L = a_x/2$$

Problem 5 — Dry Soil at 4 Frequencies (Loss Tangent Classification) 24 pts

Setup: $\epsilon_r = 2.5$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m.

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

ϵ''/ϵ'	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

$\epsilon' = \epsilon_r \epsilon_0$; $\epsilon'' = \sigma/\omega$; nonmag.: $\mu = \mu_0$; free space: $\epsilon_r = \mu_r = 1$.

LOW-LOSS MEDIUM ($\epsilon''/\epsilon' \ll 1$)

$$\begin{aligned} \alpha &\approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon'}} = \frac{\sigma \eta}{2} \text{ (Np/m)} \\ \beta &\approx \omega \sqrt{\mu \epsilon'} = \frac{\omega \sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad u_p \approx \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s)} \\ \eta_c &\approx \sqrt{\frac{\mu}{\epsilon'}} = \eta \text{ (}\Omega\text{, } \angle \approx 0) \end{aligned}$$

ANY MEDIUM — EXACT ($X = \sqrt{1 + (\epsilon''/\epsilon')^2}$)

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X-1} \text{ Np/m}, \quad \beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X+1} \text{ rad/m}$$

$$\lambda = \frac{2\pi}{\beta} = \frac{u_p}{f} \text{ (m)}, \quad u_p = \frac{\omega}{\beta} \text{ (m/s)}$$

$$\eta_c = \frac{\eta}{X^{1/2}} e^{j\theta_\eta} \text{ (}\Omega\text{)}, \quad \theta_\eta = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'} \text{ (rad)}$$

Rect: $\eta_c = |\eta_c| \cos \theta_\eta + j|\eta_c| \sin \theta_\eta$, $|\eta_c| = \eta/X^{1/2}$
 $\eta = \eta_0/\sqrt{\epsilon_r}$; $\epsilon' = \epsilon_r \epsilon_0$.

Given $\tilde{\mathbf{H}} \propto e^{-\alpha y} e^{-j\beta y}$: $\gamma = \alpha + j\beta$; $\eta_c = j\omega \mu_0/\gamma = \omega \mu_0(\beta + j\alpha)/(\alpha^2 + \beta^2)$

GOOD CONDUCTOR ($\epsilon''/\epsilon' \gg 1$)

$$\begin{aligned} \alpha &\approx \beta \approx \sqrt{\pi f \mu \sigma} \text{ (Np/m, rad/m)} \\ u_p &\approx \sqrt{4\pi f / \mu \sigma} \text{ (m/s)}, \quad \lambda \approx u_p/f \text{ (m)} \\ \eta_c &\approx (1+j) \sqrt{\frac{\pi f \mu}{\sigma}} \approx (1+j) \frac{\alpha}{\sigma} \text{ (}\Omega\text{)} \end{aligned}$$

Compute the loss tangent ratio:

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0} = \frac{\sigma/(2\pi \epsilon_r \epsilon_0)}{f} = \frac{7.193 \times 10^5 \text{ Hz}}{f}$$

f	$\varepsilon''/\varepsilon'$	Class	Use
60 Hz	1.20×10^4	Good conductor	$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$
1 kHz	719.3	Good conductor	GC approx
1 MHz	0.719	Quasi-conductor	Exact eqs
1 GHz	7.19×10^{-4}	Low-loss	LD approx

(a) **60 Hz** (good conductor):

$$\alpha \approx \beta = \sqrt{\pi(60)(4\pi \times 10^{-7})(10^{-4})} = 4.87 \times 10^{-6} \text{ Np/m, rad/m}$$

$$\lambda = 2\pi/\beta = 1.29 \times 10^6 \text{ m, } u_p = \sqrt{4\pi f/\mu\sigma} = 7.74 \times 10^7 \text{ m/s}$$

$$\eta_c \approx (1+j)\alpha/\sigma = (1+j)(0.0487) \Omega$$

(b) **1 kHz** (good conductor): same formulas, scale by $\sqrt{1000/60} \approx 4.08\times$:

$$\alpha \approx \beta = 1.99 \times 10^{-5} \text{ Np/m, } \lambda \approx 3.16 \times 10^5 \text{ m}$$

(c) **1 MHz** (quasi-conductor): use exact $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2} = \sqrt{1 + 0.719^2} = 1.231$:

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon'}{2}} \sqrt{X-1} \text{ Np/m, } \beta = \omega \sqrt{\frac{\mu\varepsilon'}{2}} \sqrt{X+1} \text{ rad/m}$$

$$\eta_c = \frac{\eta}{\sqrt{X}} e^{j\theta_\eta}, \quad \theta_\eta = \frac{1}{2} \arctan(0.719) = 17.7^\circ$$

(d) **1 GHz** (low-loss dielectric):

$$\alpha \approx \frac{\sigma}{2} \sqrt{\mu/\varepsilon'} = \frac{10^{-4}}{2} \cdot \frac{377}{\sqrt{2.5}} = 0.0119 \text{ Np/m}$$

$$\beta \approx \omega \sqrt{\mu\varepsilon'} = \frac{2\pi \times 10^9 \sqrt{2.5}}{3 \times 10^8} = 33.1 \text{ rad/m}$$

$$\lambda \approx 2\pi/\beta = 0.190 \text{ m, } u_p \approx c/\sqrt{\varepsilon_r} = 1.90 \times 10^8 \text{ m/s, } \eta_c \approx 238.4 \Omega$$

Problem 6 — Phasor → Time Domain (Lossy)

10 pts

Setup: 300 MHz, nonmagnetic lossy, $\tilde{\mathbf{H}} = (\hat{x} - j4\hat{z})e^{-2y}e^{-j9y}$ A/m.

PHASOR ↔ TIME DOMAIN		
Phasor coeff	$\text{Re}\{(\cdot)e^{j\omega t}e^{-j\beta z}\}$	Result
A	$\text{Re}\{Ae^{j(\omega t - \beta z)}\}$	$A \cos(\omega t - \beta z)$
$-jA$	$\text{Re}\{Ae^{j(\omega t - \beta z - \pi/2)}\}$	$A \sin(\omega t - \beta z)$
$+jA$	$\text{Re}\{Ae^{j(\omega t - \beta z + \pi/2)}\}$	$-A \sin(\omega t - \beta z)$
$-A$	$\text{Re}\{Ae^{j(\omega t - \beta z \pm \pi)}\}$	$-A \cos(\omega t - \beta z)$
$Ae^{j\phi}$	$\text{Re}\{Ae^{j(\omega t - \beta z + \phi)}\}$	$A \cos(\omega t - \beta z + \phi)$
$A + jB$	$\text{Re}\{ Z e^{j(\omega t - \beta z + \theta)}\}$	$ Z \cos(\omega t - \beta z + \theta)$
$ Z = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check quadrant)		

$\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION					
$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$-\hat{z}$	$\hat{x}$	$-\hat{y}$	$-\hat{z}$	$\hat{y}$	$+\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{y}$	$\hat{z}$	$-\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$

Neg. $\hat{E}$: flip $\hat{H}$ sign. $\hat{k} \times \hat{k} = \mathbf{0}$; any $\hat{a} \times \hat{a} = \mathbf{0}$.
 Cyclic: $\hat{x} \times \hat{y} = \hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$, $\hat{z} \times \hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

Read off: $\omega = 6\pi \times 10^8$ rad/s, $\alpha = 2$ Np/m, $\beta = 9$ rad/m, $\hat{k} = \hat{y}$.

Time-domain $\mathbf{H}$ using the phasor table ($-j \rightarrow \sin$):

$$\mathbf{H}(y, t) = \left[\cos(\omega t - 9y) \hat{x} - 4 \cos(\omega t - 9y + \frac{\pi}{2}) \hat{z} \right] e^{-2y} \text{ A/m}$$

Find $\tilde{\mathbf{E}}$ using $\tilde{\mathbf{E}} = -\eta_c(\hat{k} \times \tilde{\mathbf{H}})$, but for lossy we need η_c first. Compute loss tangent:

$$\epsilon''/\epsilon' = (\alpha\beta \cdot 2)/(\omega^2\mu\epsilon') \quad (\text{implicit from } \alpha, \beta)$$

Or compute $\eta_c = j\omega\mu_0/\gamma$ with $\gamma = \alpha + j\beta = 2 + j9$:

$$\eta_c = \frac{j(6\pi \times 10^8)(4\pi \times 10^{-7})}{2 + j9} = \frac{j(753.98)}{2 + j9} = \frac{j753.98(2 - j9)}{4 + 81} = \frac{6785.8 + j1507.9}{85} \approx 79.8 + j17.7 \Omega$$

Then $\hat{y} \times \hat{x} = -\hat{z}$, $\hat{y} \times \hat{z} = \hat{x}$:

$$\tilde{\mathbf{E}} = -\eta_c(\hat{y} \times (\hat{x} - j4\hat{z}))e^{-2y}e^{-j9y} = -\eta_c(-\hat{z} - j4\hat{x})e^{-2y}e^{-j9y}$$

$$\tilde{\mathbf{E}} = \eta_c(j4\hat{x} + \hat{z})e^{-2y}e^{-j9y} \text{ V/m}$$

Time domain:

$$\mathbf{E}(y, t) = |\eta_c| \left[4 \cos(\omega t - 9y + \pi/2 + \theta_\eta) \hat{x} + \cos(\omega t - 9y + \theta_\eta) \hat{z} \right] e^{-2y}$$

where $|\eta_c| \approx 81.8$, $\theta_\eta = \arctan(17.7/79.8) \approx 12.5^\circ$.

Problem 7 — Direction of Travel and Poynting

10 pts

Setup: $\mathbf{E} = 3 \cos(\pi \times 10^7 t + kx) \hat{y} - 2 \cos(\pi \times 10^7 t + kx) \hat{z}$ V/m, nonmagnetic $\epsilon_r = 9$.

DIRECTION OF PROPAGATION			
Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$
<i>Minus before k-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ - direction.</i>			

AVERAGE POWER DENSITY (POYNTING)	
General	W/m^2
$\mathbf{S}_{av} = \frac{1}{2} \text{Re}\{\tilde{\mathbf{E}} \times \tilde{\mathbf{H}}^*\}$	
Lossless shortcut $ E_0 ^2 = a_x^2 + a_y^2 + a_z^2$	
$\mathbf{S}_{av} = \frac{ E_0 ^2}{2\eta} \hat{k} \text{ (W/m}^2\text{)}$	
Lossy medium $\eta_c = \eta_c e^{j\theta_\eta}$; wave in $+\hat{z}$	
$\mathbf{S}_{av} = \frac{ \tilde{\mathbf{E}}(0) ^2}{2 \eta_c } e^{-2\alpha z} \cos\theta_\eta \hat{z} \text{ (W/m}^2\text{)}$	
<i>Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative x-component.</i>	

Direction from the table: argument is $\omega t + kx$ (plus sign $\Rightarrow$ wave travels in $-\hat{x}$).

$$\hat{k} = -\hat{x}$$

No $e^{-\alpha x}$ attenuation factor, so the medium is effectively lossless here.

Intrinsic impedance:

$$\eta = \eta_0 / \sqrt{\epsilon_r} = 377/3 \approx 125.7 \, \Omega = 40\pi \, \Omega$$

Poynting (lossless shortcut): $|E_0|^2 = 3^2 + 2^2 = 13$.

$$\mathbf{S}_{av} = \frac{|E_0|^2}{2\eta} \hat{k} = \frac{13}{80\pi} (-\hat{x}) \text{ W/m}^2 \approx -51.7 \text{ mW/m}^2 \hat{x}$$

$$\mathbf{S}_{av} \approx -51.7 \text{ mW/m}^2 \hat{x}$$